Prime Certificate:
The Mod-6 Wheel, Base 60, and the Totient

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Abstract

This note records the structural chain connecting a period-6 sawtooth, the base-60 numeral inheritance, wheel factorization, and Euler's totient. The central result is that a single function, φ, plays two roles: it measures the *width of a prime-sieving wheel* and it encodes the *primality of an individual integer*.

1 Background: the mod-6 wheel

Every integer occupies one of six lanes modulo 6. Lanes $0, 2, 4$ are even and lane 3 is divisible by 3, so beyond the base primes 2 and 3 only lanes 1 and 5 can hold primes:

$$p > 3 \implies p \equiv 1 \text{ or } 5 \pmod 6, \qquad \text{i.e.} \qquad p = 6m \pm 1.$$

These two survivors are exactly the fold (reset) positions of the two sawtooth waves

$$B = \big((x - \tfrac{6}{8}) \bmod 6\big) \cdot 2, \qquad D = \big((x + 2 - \tfrac{6}{8}) \bmod 6\big) \cdot 2,$$

which reset at $x \bmod 6 = 1$ and $x \bmod 6 = 5$ respectively.

2 Base-60 inheritance

Base 60 factors as $60 = 6 \times 10$: a mod-6 "tens" wheel geared to a base-10 "units" wheel. We compute in base 10 but time and angle keep the base-60 rollover, whose reset is precisely the $x \bmod 6 = 5$ event of the wheel above. The sieving power of 60 lives entirely in its squarefree kernel $30 = 2 \cdot 3 \cdot 5$.

Details: The Totient Relation

Euler's totient φ is the hinge of this entire construction. It appears in *two* distinct roles: once attached to the wheel modulus M, and once attached to the integer N under test.

Definition 1 (Totient)**.** For a positive integer n,

$$\varphi(n) \;=\; \big|\{\, k : 1 \le k \le n,\ \gcd(k, n) = 1 \,\}\big|,$$

the count of integers up to n that are coprime to n.

Role 1 — $\varphi(M)$ is the wheel width

For a wheel with modulus M, the spokes are exactly the residues coprime to M, so the number of spokes is $\varphi(M)$ and the surviving density of the number line is $\varphi(M)/M$. This density is governed by the multiplicative product formula.

Theorem 1 (Euler product formula)**.**

$$\varphi(M) \;=\; M \prod_{p \mid M} \Big(1 - \frac{1}{p}\Big),$$

the product ranging over the distinct *primes dividing M. Consequently the wheel density is*

$$\frac{\varphi(M)}{M} \;=\; \prod_{p \mid M} \Big(1 - \frac{1}{p}\Big).$$

Each factor $(1 - 1/p)$ is one prime's worth of lanes struck out. This one equation explains every empirical observation about the wheels:

M	$\varphi(M)$	$\varphi(M)/M$	$\prod(1-1/p)$	distinct primes of M
6	2	0.3333	0.3333	$\{2,3\}$
30	8	0.2667	0.2667	$\{2,3,5\}$
60	16	0.2667	0.2667	$\{2,3,5\}$ (no gain over 30)
210	48	0.2286	0.2286	$\{2,3,5,7\}$
2310	480	0.2078	0.2078	$\{2,3,5,7,11\}$

Two consequences follow immediately:

- 60 **gains nothing over** 30**:** they share the distinct primes $\{2, 3, 5\}$, so the product—and hence the density—is identical. The extra factor of 2 in $60 = 2^2 \cdot 3 \cdot 5$ never enters the product, which counts only *distinct* primes.
- 210 **does gain:** it introduces a genuinely new factor $(1 - 1/7)$, lowering the density from 0.2667 to 0.2286.

Role 2 — $\varphi(N)$ is a primality signature

For the integer N passed to the scan, the totient encodes primality directly:

$$N \text{ is prime} \iff \varphi(N) = N - 1, \qquad\qquad \varphi(p^k) = p^k - p^{k-1}.$$

A prime shares a factor with nothing below it, so all $N - 1$ smaller integers are coprime—the maximal possible totient. Every composite satisfies $\varphi(N) < N - 1$.

The bridge — Euler's theorem

The two roles are joined by Euler's theorem, which also underlies the fast primality tests (Fermat, Miller–Rabin) that outperform trial division for large N.

Theorem 2 (Euler)**.** *If* $\gcd(a, N) = 1$ *then*

$$a^{\varphi(N)} \equiv 1 \pmod{N}.$$

For prime p this specialises, via $\varphi(p) = p - 1$, to Fermat's little theorem $a^{p-1} \equiv 1 \pmod{p}$.

Miller–Rabin checks whether N *violates* this congruence: a violation certifies compositeness with a single modular exponentiation, whereas the wheel must trial-divide up to $\sqrt{N}$.

Summary

$$\boxed{\underbrace{\varphi(M)}_{\text{sieve width}} = M\prod_{p\mid M}\Bigl(1-\tfrac{1}{p}\Bigr) \qquad \text{and} \qquad \underbrace{\varphi(N)=N-1}_{\text{primality signature}} \iff N \text{ prime},}$$

the same function serving once as a wheel's width and once as an integer's primality signature, bridged by $a^{\varphi(N)} \equiv 1 \pmod{N}$.

3 Reference implementation

The mod-210 wheel ($210 = 2\cdot3\cdot5\cdot7$, $\varphi(210) = 48$ spokes) used as a trial-division primality scanner:

```python
from math import gcd, isqrt

WHEEL210 = [r for r in range(210) if gcd(r, 210) == 1]    # 48 spokes

def wheel_scan_210(N):
    """Smallest factor of N via a mod-210 wheel, or None if N is prime."""
    if N < 2:
        return None
    for p in (2, 3, 5, 7):                  # base primes the wheel skips
        if N == p:  return None
        if N % p == 0: return p
    limit = isqrt(N)
    m = 0
    while 210*m + 1 <= limit:
        for r in WHEEL210:
            n = 210*m + r                   # actual candidate = block offset + spoke
            if n == 1:                      # skip trivial divisor
                continue
            if n > limit:
                break
            if N % n == 0:
                return n                    # real factor -> composite
        m += 1
    return None                             # nothing up to sqrt(N) -> prime

def is_prime(N):
    return N >= 2 and wheel_scan_210(N) is None
```